WORKED SOLUTION

STEP I 1987, Question 12

Mechanics • Collision

STEP I 1987, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q12

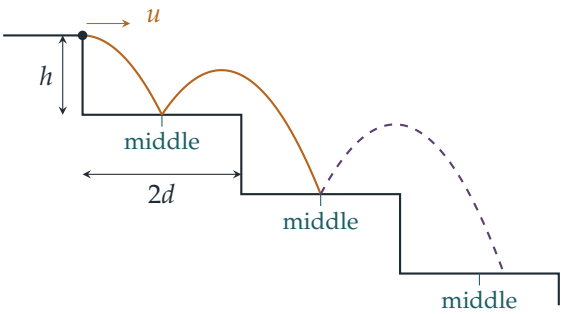
The key idea

Between bounces the particle is a projectile, and (with smooth steps) a bounce reverses and scales only the vertical component of velocity, so the horizontal speed never changes. That means the horizontal distances fix the flight times, and the flight times together with the drop of one step fix the vertical speeds. Everything can be measured in terms of the time of the first fall, which keeps the algebra tiny.

Solution

Model

Treat the particle as a point mass, ignore air resistance, and take the steps to be smooth and horizontal, so an impact leaves the horizontal velocity unchanged and multiplies the vertical velocity by $-e$. Let the particle be kicked with horizontal speed u, and let t_1 be the time of the first flight.



The solid arcs are the first two flights; the dashed arc is what actually happens next (see the final part).

The coefficient of restitution

First flight. The particle starts with no vertical velocity, falls h and travels d horizontally (half a step width, to the middle of the first step). So

$$h = \tfrac{1}{2}gt_1^2, \qquad d = ut_1.$$

It strikes the first step with downward vertical speed gt_1, and leaves it moving upwards with vertical speed egt_1, still with horizontal speed u.

Second flight. From the middle of the first step to the middle of the second is a horizontal distance of $2d$. Since the horizontal speed is still $u = d/t_1$, this flight lasts

$$T = \frac{2d}{u} = 2t_1.$$

In that time the particle must end up a height h lower. Taking upwards as positive,

$$\begin{aligned} -h &= (egt_1)(2t_1) - \tfrac{1}{2}g(2t_1)^2 \\ &= (2e-2)\,gt_1^2 \\ &= (2e-2)(2h), \end{aligned}$$

using $gt_1^2 = 2h$. Dividing by h gives $-1 = 4e - 4$, so

$$e = \tfrac{3}{4}.$$

(On this flight the particle is a height $egt_1 \cdot t_1 - \frac{1}{2}gt_1^2 = \frac{1}{2}h$ above the first step when it passes over that step's edge, so it does clear the edge.)

Can it keep going?

> **Ask yourself** What would every later flight have to look like if the pattern continued?

Every flight from the middle of one step to the middle of the next covers $2d$ horizontally at speed u, so each lasts $2t_1$, and each must drop exactly h. The calculation above then shows that the upward vertical speed w just after each bounce must satisfy

$$-h = w(2t_1) - 2gt_1^2 = 2wt_1 - 4h, \qquad \text{so} \qquad w = \frac{3h}{2t_1} = \tfrac{3}{4}gt_1.$$

So after every bounce the particle must leave with vertical speed $\frac{3}{4}gt_1$; since $e = \frac{3}{4}$, it must arrive at every step with vertical speed exactly gt_1.

That holds at the first step, but not at the second. Arriving there after a flight of time $2t_1$, its vertical velocity is

$$\tfrac{3}{4}gt_1 - g(2t_1) = -\tfrac{5}{4}gt_1,$$

so it hits the second step with downward speed $\frac{5}{4}gt_1$, not gt_1, and rebounds with upward speed $\frac{3}{4} \cdot \frac{5}{4}gt_1 = \frac{15}{16}gt_1 > \frac{3}{4}gt_1$. So it is **not possible**: the particle cannot hit the

middle of every step. In fact it overshoots the middle of the third step. If the third flight lasts st_1, then

$$-h = \tfrac{15}{16}gt_1(st_1) - \tfrac{1}{2}gs^2t_1^2 \quad\Longrightarrow\quad s^2 - \tfrac{15}{8}s - 1 = 0 \quad\Longrightarrow\quad s = \frac{15+\sqrt{481}}{16} \approx 2.31,$$

so it travels about $2.31d$ horizontally and lands on the third step about $0.31d$ beyond its middle (still on that step, since the far edge is $3d$ away).

Checks

- $e = \frac{3}{4}$ does not depend on g, h or d, as it shouldn't: scaling the staircase changes nothing about the shape of the path.
- A numerical simulation with $g = 9.8$, $h = d = 1$ and $e = 0.75$ lands at $x = 1.000$ and $x = 3.000$ (the middles of the first two steps) and then at $x \approx 5.308$, against $3 + 2.308$ predicted above.
- Without a bounce ($e = 0$) the second flight would fall $4h$ in time $2t_1$; the rebound must make up $3h$ of that, giving $2t_1 \cdot w = 3h$, as found.

Common slips

- Taking the first horizontal distance as $2d$ rather than d: the particle starts at the edge and lands in the middle of the first step.
- Changing the horizontal speed at the bounce. On a smooth surface the impulse is vertical, so only the vertical component changes.
- Concluding "yes" because each flight looks identical: the arrival speeds grow, so the rebounds grow too.

Beyond STEP

A bouncing particle is a simple example of a *discrete dynamical system*: the state just after one bounce determines the state after the next through a fixed map, and asking whether the motion can repeat is asking for a fixed point of that map. Here the fixed point (rebound $\frac{6}{7}gt_1$, which would land only $\frac{4}{7}h$ lower) is incompatible with the staircase, so no periodic motion fits. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and impulse carefully.

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